

User Guide for the Thermal Image Mosaic Application

The Thermal Image Mosaic is a cross-platform graphical application that allows users to visually compare, align, and combine thermal and RGB images on a unified working canvas. The following steps explain how to install and use the application, from setup to exporting the final image.

1. System Requirements

Python version: 3.10 recommended (tested with 3.10.x)

Python can be downloaded from: <https://python.org>

Supported operating systems:

- Windows 10 / 11
- Linux (e.g., Ubuntu 22.04, Linux Mint)
- macOS 11 “Big Sur” or later

2. Downloading the Project from GitHub

<https://github.com/norbistopgit/ThermalImageMosaic>

- Click on “Code” → “Download ZIP”
- Extract the ZIP file, e.g., to a folder named ThermalImageMosaic

3. Folder Structure

The application expects the following folder structure: ThermalImageMosaic/

ThermalImageMosaic/

├─ ThermalImageMosaic.py

├─ main_window.py

├─ constants.py

├─ image_loader.py

├─ save_images.py

├─ images/

| └─ rgb/

| └─ termal/

├─ help_documentation.pdf

└─ icon.png

4. Installing Dependencies

To run the program, install the following external Python libraries:

- `pip install pillow`
- `pip install tkinterdnd2`

For Linux systems, you may also need to install additional packages:

- `sudo apt install python3-tk python3-dev`

5. Running the Application

Use the terminal or command prompt:

- `python ThermalImageMosaic.py`

On Windows, you can also double-click the .py file if it's associated with the Python interpreter.

6. Preparing Images

Within the `images/` folder, the following subfolders are used:

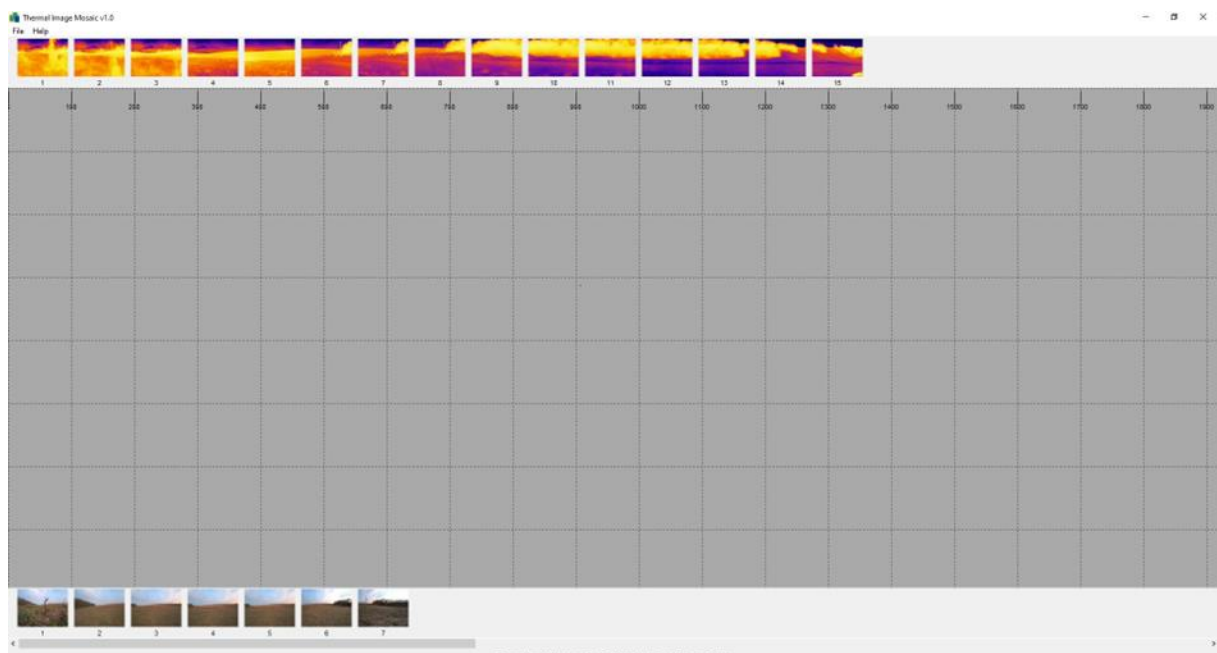
- `images/rgb`: for RGB images (e.g., from regular drone cameras)
- `images/thermal`: for thermal camera images

Images placed in these folders will automatically be detected and displayed as thumbnails in the application upon launch.

7. Thumbnails and Drag & Drop

Clicking on any thumbnail places the image onto the canvas.

You can also drag & drop images directly from your file explorer into the canvas. Supported formats include .jpg, .png, and .bmp.



8. Editing the Canvas

Images placed on the canvas can be freely moved using the mouse.

They are automatically scaled while maintaining aspect ratio and include an alpha channel for transparency. This allows you to overlay and align images visually.

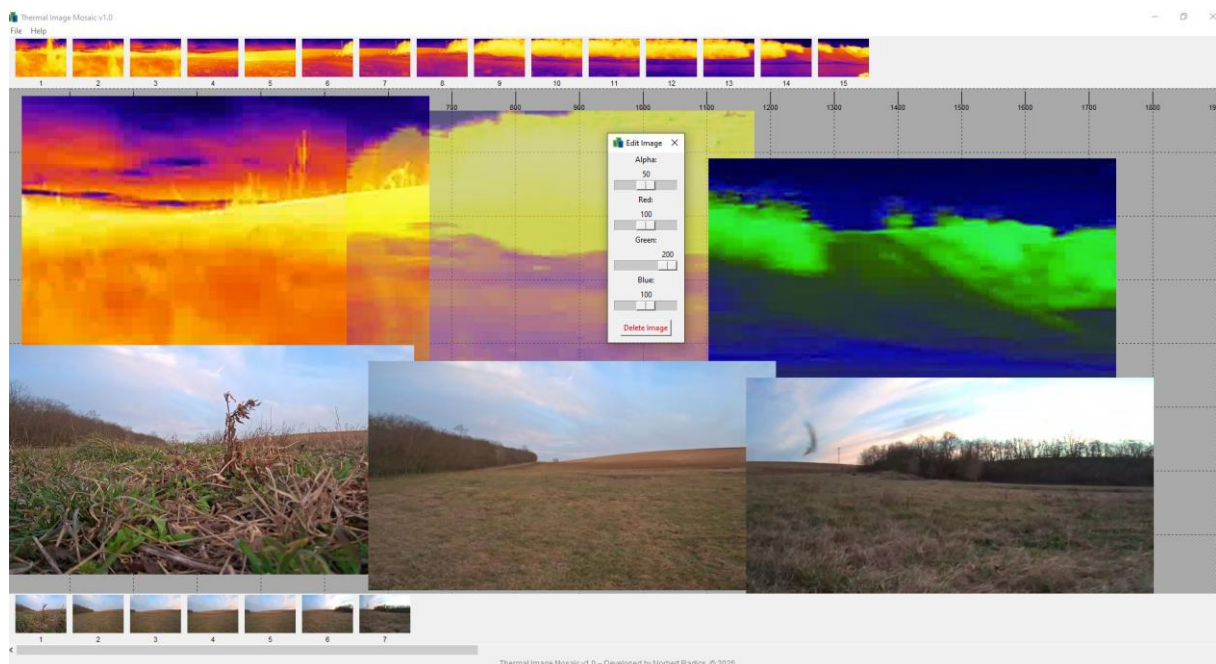
The alpha channel helps in comparing thermal and RGB data by making overlapping images semi-transparent.

9. Editing Images: RGB Channels and Transparency

On macOS, use a two-finger tap on the trackpad to select an image (a red border will appear around it), then click the “Edit Image” button to open the adjustment panel where you can fine-tune the RGB channels and transparency (Alpha).

Available settings:

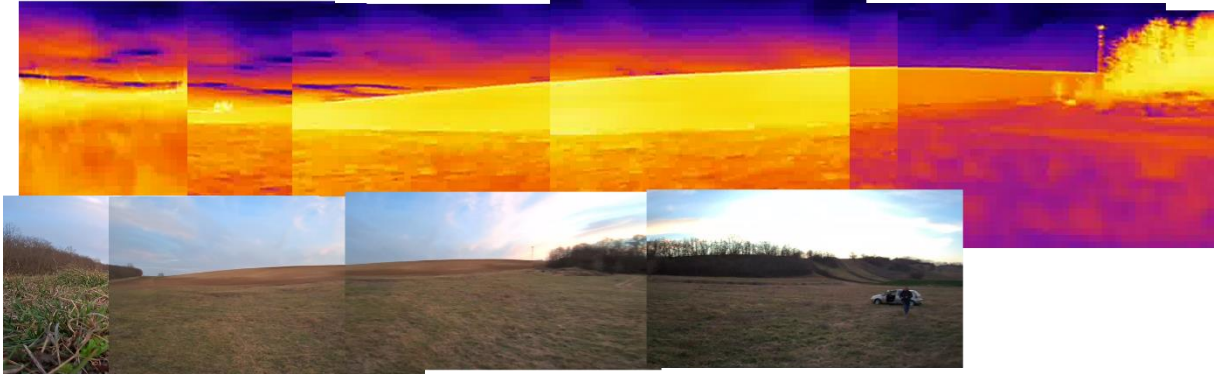
- Alpha (Transparency): Adjustable from 0% to 100%
- R, G, B Channels: Adjustable from 0% to 200% to fine-tune image tones and contrast
- A Delete Image button is available to remove the image from the canvas.



All adjustments are applied in real-time and non-destructively – changes can always be reversed.

10. Saving Options – File Menu

- Export Canvas View: Saves a screenshot-like image of the current canvas in PNG, JPG, or BMP format, including all positioned and edited images as displayed.
- Exit: Closes the application.



11. Help and Information – Help Menu

- Open Help: Opens the help_documentation.pdf file with detailed user instructions.
- About: Displays program information, version, author, and license details.